

Tagging Hosts with deployment.environment and Custom Attributes

Use the OpenTelemetry deployment.environment attribute and custom resource attributes to tag your host, via the Collector's Resource processor — the method Splunk's official docs recommend.

Prerequisite

Part 1 completed — Collector running as a Windows service

Objective & Prerequisite

Objective

Tag hosts using the standardized OpenTelemetry deployment.environment attribute — not a generic "environment" tag — plus custom resource attributes, applied through the Collector's Resource processor.

Reference documentation

- Use tags or attributes in OpenTelemetry
- Resource processor documentation

Prerequisite

Completed lab-otel-collector-windows-part1.md:

- Collector installed and running as a Windows service
- Host metrics visible in Infrastructure → Hosts

Estimated time

15 minutes

Why deployment.environment Specifically

deployment.environment is a standardized OpenTelemetry attribute — not a free-form tag. Splunk Observability Cloud uses this exact name to power Related Content links between Infrastructure and APM.

Generic tag

```
environment: staging  
env: staging
```

Custom keys aren't recognized by Splunk's Related Content linking — Infrastructure and APM stay disconnected.

Standardized attribute

```
deployment.environment: staging
```

The exact OTel-standard key. Splunk Observability Cloud recognizes it and links host and service views automatically.

Open the Config & Add a Resource Processor

1 Open the agent config

```
notepad "C:\ProgramData\Splunk\OpenTelemetry Collector\agent_config.yaml"
```

2 Add a resource processor — find processors: and add a resource: block

```
processors:
  resourcedetection:
    detectors: [system]
    override: true

  resource:
    attributes:
      - key: deployment.environment
        value: "staging"
        action: upsert
      - key: team
        value: "web"
        action: upsert
```

STEP 3

Wire resource Into the Pipelines

Under service: → pipelines:, list resource in the processors: array for the pipelines you want tagged — typically both metrics and traces.

```
service:
  pipelines:
    metrics:
      receivers: [hostmetrics, windowsperfcounters]
      processors: [resourcedetection, resource, batch]
      exporters: [signalfx]

    traces:
      receivers: [otlp]
      processors: [resourcedetection, resource, batch]
      exporters: [sapm]
```

Processor order matters

resourcedetection runs first and fills in defaults (like hostname). resource runs second and applies your upsert / insert / delete actions on top.

upsert

Add if missing, overwrite if present

insert

Only if the key doesn't already exist

delete

Remove a key

Save, Restart, and Check for Errors

4 Save and restart the service

```
Restart-Service splunk-otel-collector  
Get-Service splunk-otel-collector
```

Expected result: **Status shows Running**

5 Check for config errors

```
Get-Content "C:\ProgramData\Splunk\OpenTelemetry Collector\logs\otelcol.log" -Tail 30
```

Tip: A YAML indentation mistake is the most common failure here — the log will point at the offending line if the service won't come back up.

Validate in the UI

1

Open your host

Go to Infrastructure → Hosts in Splunk Observability Cloud and open your host.

2

Check metadata

Check the Metadata / Properties panel for `deployment.environment: staging` and `team: web`.

3

Confirm APM linking

In APM, confirm a service on this host shows a Related Content link back to its Infrastructure host — appears only when `deployment.environment` is set correctly.

4

Search the filter bar

On any chart or host list, search `deployment.environment:staging` to confirm the host is discoverable.

CHECKPOINT

Part 2 (Exercise 7) Checkpoint



resource processor added with deployment.environment (upsert)



resource processor included in metrics/traces pipelines



Collector restarted with no errors in otelcol.log



deployment.environment and custom attributes visible on host in UI

Next: Continue with Exercise 8 (alerting) in lab-otel-collector-windows-part2.md